

MZ2POL-04: Policies and Boundaries

Series: MZ2POL — Management Zone to Policy Migration | **Notebook:** 5 of 9 |
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Overview

This notebook provides a comprehensive guide to **IAM Policies** and **Policy Boundaries** - the two components that together replace Management Zone access control. Policies define **WHAT** users can do, while boundaries define **WHERE** those permissions apply.

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Prerequisites

- Completed MZ2POL-01 through MZ2POL-03
- Access to Dynatrace Account Management
- Policy Management permissions

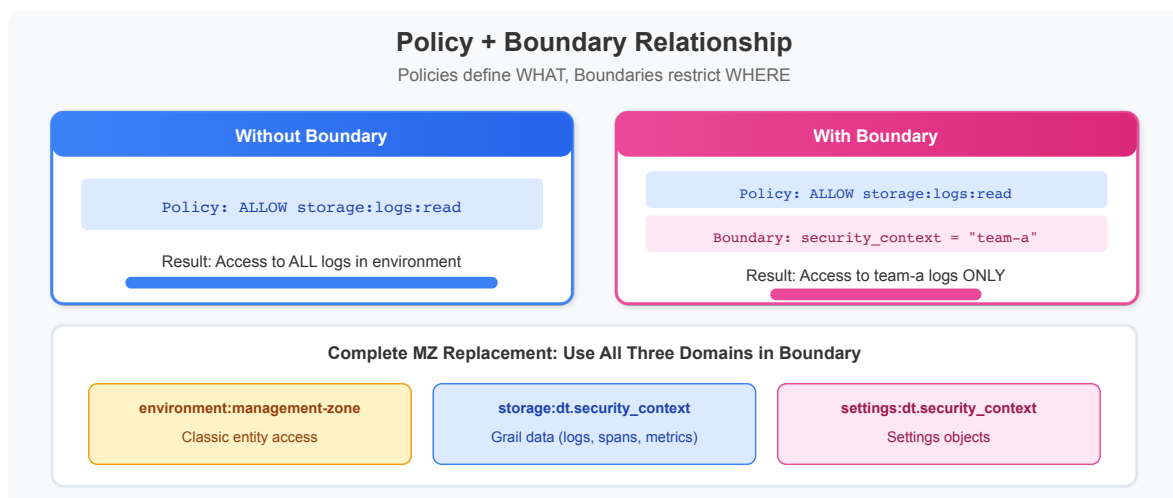
Learning Objectives

By the end of this notebook, you will:

1. Understand policy statement syntax and structure
2. Know the default policies and when to use them
3. Understand boundary syntax and how boundaries restrict policies
4. Be able to map MZ access patterns to policy + boundary combinations
5. Know the recommended group-policy-boundary structure for SAML/AD integration

1. Understanding the Relationship

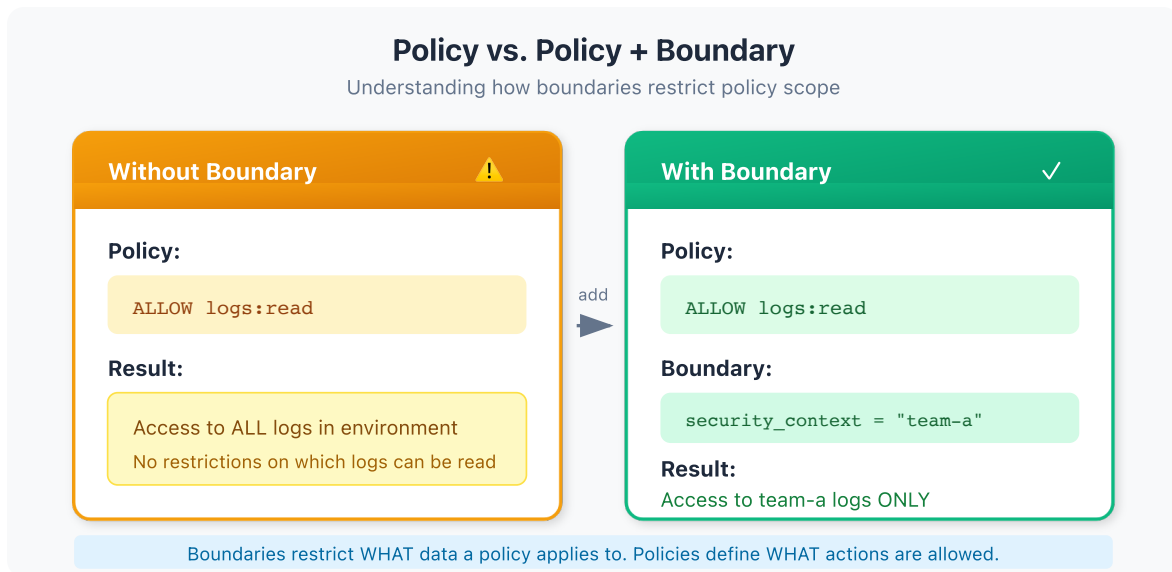
Policies vs Boundaries



Component	Purpose	Analogy
Policy	Defines WHAT actions are allowed	"You can read logs"

Component	Purpose	Analogy
Boundary	Defines WHERE policies apply	"...but only for team-frontend"

How They Work Together



Key Characteristics

Policies:

- Define permissions (ALLOW statements)
- Can include conditions (WHERE clauses)
- Default policies cover most use cases

Boundaries:

- Separate from policies (decoupled)
 - Reusable across multiple policies
 - Can only narrow permissions, never expand
 - Optional - policies work without them (full scope)
-

2. Policy Fundamentals

Policy Statement Structure

```
<effect> <service>:<resource>:<action> [WHERE  
<condition>]
```

Component	Description	Examples
Effect	Permission type	ALLOW or DENY (a DENY always overrules a matching ALLOW)
Service	Dynatrace service	storage, settings, app
Resource	Resource type	logs, buckets, objects
Action	Operation	read, write, delete
Condition	Optional filter	WHERE field = "value"

Effects: Authorization is **implicit-deny** — anything not ALLOW ed is denied. Compose access from ALLOW statements and simply omit what you don't want to grant. Reserve explicit DENY for narrowing a broad ALLOW ; a DENY always wins over any ALLOW , regardless of order.

Combining conditions: A policy WHERE clause supports **AND** (e.g. WHERE settings:schemaId = "... " AND storage:bucket-name = "logs_prod"). It does **not** support **OR** — to express OR, write multiple ALLOW statements.

Basic Policy Statements

```
// Grant read access to all logs
ALLOW storage:logs:read

// Grant read/write to buckets
ALLOW storage:buckets:read
ALLOW storage:buckets:write

// Grant all actions on a resource
ALLOW storage:logs:*
```

Common Services and Resources

Service	Resources	Common Actions
storage	logs , spans , events , metrics , buckets	read , write , delete
settings	objects , schemas	read , write , delete
app	apps , functions	run , install
automation	workflows	read , write , run
document	documents	read , write , delete , share

3. Default Policies

Dynatrace provides **built-in (default) IAM policies** that cover common access patterns. They are managed by Dynatrace (not editable) and auto-adjust as the platform evolves.

Naming note: The classic Gen2 roles "Dynatrace Viewer / Standard User / Professional User / Admin User" and "Data Viewer / Data Editor" are **not** current IAM policy names — they were RBAC roles. The names below are the current built-in IAM policies (see the [default policies reference](#)).

Platform Access Policies

Policy	Use Case	Grants
Standard User	Regular users	Access the environment and run Dynatrace Apps
Pro User	Power users	Build, deploy, and run apps + automated workflows
Admin User	Administrators	Administrative access across all Platform Services

There is **no built-in "Viewer" policy** — compose read-only access from a platform policy plus the granular data-read policies below.

Data Access Policies

Policy	Use Case
All Grail data read access	Unrestricted read across all Grail tables and buckets
Storage Default Monitoring Read	Read on the default monitoring buckets; auto-extends as new tables are added
Read Logs / Read Spans / Read Metrics / Read Events / Read Entities	Granular read, one policy per signal type
Settings Reader / Settings Writer	Read / read-write of Settings 2.0 objects

Environment Role Policies (Classic surface)

Policy	Use Case
Environment role - Access environment	Viewer-level Classic environment access
Environment role - Change monitoring settings	Edit Classic monitoring settings

Choosing the Right Built-in Policies

User Type	→ Platform Policy	+ Data / Settings Policies
Executives/Viewers	→ Standard User	+ Read Logs / Read Metrics (or All Grail data read access)
Developers	→ Standard User	+ Read Logs / Read Spans / Read Metrics
SRE/Operations	→ Pro User	+ All Grail data read access +
Settings Writer		
Platform Admins	→ Admin User	+ (admin policies as needed)

4. Condition and Boundary Syntax

Policies and boundaries share the same condition syntax.

Supported Operators

Operator	Description	Example
<code>=</code>	Exact match	<code>field = "value"</code>
<code>!=</code>	Not equal	<code>field != "value"</code>
<code>startsWith</code>	Prefix match	<code>field startsWith "prefix"</code>
<code>in</code>	Value in list	<code>field in ("a", "b", "c")</code>
<code>IN</code>	Value in list (alternative)	<code>field IN ("a", "b")</code>
<code>MATCH</code>	Wildcard pattern match (storage domain only)	<code>field MATCH ('team-*')</code>

Supported Fields

Field	Description	Used In
<code>environment</code>	Environment restrictions	Boundaries

Field	Description	Used In
<code>environment:name</code>	Environment by name	Boundaries
<code>environment:management-zone</code>	MZ-based (transitional)	Boundaries
<code>storage:dt.security_context</code>	Security context	Both
<code>storage:bucket-name</code>	Grail bucket	Both
<code>settings:schemaId</code>	Settings schema ID	Policies
<code>settings:dt.security_context</code>	Settings security context	Boundaries

Combining Conditions

AND logic (all conditions must match):

```
ALLOW storage:logs:read
  WHERE storage:dt.security_context = "team-a"
  AND storage:bucket-name = "production_logs"
```

OR logic (multiple statements or lines):

```
// In policies: Multiple statements
ALLOW storage:logs:read WHERE storage:dt.security_context = "team-a"
ALLOW storage:logs:read WHERE storage:dt.security_context = "team-b"

// In boundaries: Each line is OR-combined
storage:dt.security_context = "team-a"
storage:dt.security_context = "team-b"
```

5. Creating Boundaries

Via UI

1. Navigate to **Account Management** → **Identity & Access Management**
2. Select **Policy Boundaries** → **Boundaries** tab
3. Click **Create boundary**

4. Enter:
- **Boundary name:** Descriptive name
 - **Boundary query:** Restriction conditions
5. Click **Save**

Boundary Examples

Team-Based Boundary:

```
Name: Frontend Team Scope
Query: storage:dt.security_context = "team-frontend"
```

Environment Boundary:

```
Name: Production Only
Query: storage:dt.security_context startsWith "prod-"
```

Multi-Team Boundary (OR logic):

```
Name: Platform Teams
Query:
storage:dt.security_context = "team-infra"
storage:dt.security_context = "team-sre"
storage:dt.security_context = "team-platform"
```

Boundary Limitations

Limitation	Description	Workaround
Max 10 lines	Only 10 conditions per boundary	Create multiple boundaries
No AND in lines	Each line is one condition	Use multiple boundaries for AND
No complex expressions	Limited to basic operators	Simplify conditions

6. Best Practice: Two Boundaries (Gen3 + Gen2 Transitional)

When restricting access to replicate a Management Zone, **don't bundle Gen2 and Gen3 conditions inside one boundary**. Split them across **two parallel policy bindings** on the same group — a Gen3 binding scoped by Gen3 conditions and a Gen2 binding scoped by the `environment:management-zone` condition. Their effective scope is the intersection.

```
# Gen3 policy binding (canonical) – Grail data + Settings 2.0
ALLOW storage:logs:read, storage:spans:read, storage:metrics:read,
      settings:objects:read, settings:objects:write
WHERE storage:dt.security_context IN ("LOB5")
      AND settings:dt.security_context IN ("LOB5");
```

The `WHERE` clause uses `AND`, not `OR` — a policy `WHERE` does not support `OR`. Each condition only constrains the permissions of its own service:
`storage:dt.security_context` scopes the `storage:*` reads, and
`settings:dt.security_context` scopes the `settings:*` actions.

```
# Gen2 policy binding (transitional) – Classic entity access via Management Zones
ALLOW environment:roles:viewer
WHERE environment:management-zone IN ("LOB5");
```

Why Split Gen2 from Gen3?

Boundary	Domain	What It Restricts	Generation
Boundary 1 (Gen3)	<code>storage:dt.security_context</code>	Grail data (logs, spans, metrics, events)	Modern, canonical
Boundary 1 (Gen3)	<code>settings:dt.security_context</code>	Settings 2.0 objects	Modern, canonical

Boundary	Domain	What It Restricts	Generation
Boundary 2 (Gen2)	<code>environment:management-zone</code>	Classic entity access (hosts, services, processes)	Transitional

Each boundary's conditions evaluate against the policy it's attached to — using `environment:management-zone` to scope a `storage:logs:read` policy is a no-op (the Gen3 policy doesn't read `environment:` conditions), and using `storage:dt.security_context` to scope an `environment:roles:viewer` policy is also a no-op. Mixing them in one boundary creates the illusion of unified scope while leaving one path wide open. Use **two parallel policy bindings** on the same group instead.

Until MZ retirement is complete, keep both bindings active so legacy entity-level paths still resolve. After retirement, the Gen2 binding is removed cleanly without touching the Gen3 one. Don't use `MATCH` on a Gen2 policy — `MATCH('*')` against `storage:entities:read` (a Gen2 surface) silently grants access to all Gen2 entities.

7. Recommended Group-Policy-Boundary Structure

For organizations using SAML/SSO with Active Directory, this structure provides clear separation of duties:

```

Group: "LOB5-Team" (SAML from Active Directory)
├─ Policy binding 1 – Gen3 (canonical):
|   Policies: Standard User (+ Read Logs / Read Spans / Read Metrics;
Settings Reader)
|   Boundary:
|       storage:dt.security_context IN ("LOB5");
|       settings:dt.security_context IN ("LOB5");
└─ Policy binding 2 – Gen2 (transitional):
    Policy: Environment role – Access environment
    Boundary:
        environment:management-zone IN ("LOB5");

```

Key Benefits

- **SAML Group:** Ties to existing AD group membership - no separate user management
- **Gen3 binding:** Grail data + Settings access scoped by `dt.security_context`
- **Gen2 binding:** Classic entity access scoped by `environment:management-zone`, retired cleanly after MZ migration

Alternative: Tighten the Platform Policy Too

For stricter isolation, bind a higher platform policy under the same Gen3 boundary:

```

Group: "LOB5-Team" (SAML from AD)
├─ Policy: Standard User
|   └─ Boundary: LOB5-Scope (Gen3)
└─ Policy: Pro User
    └─ Boundary: LOB5-Scope (Gen3)

```

Binding via UI

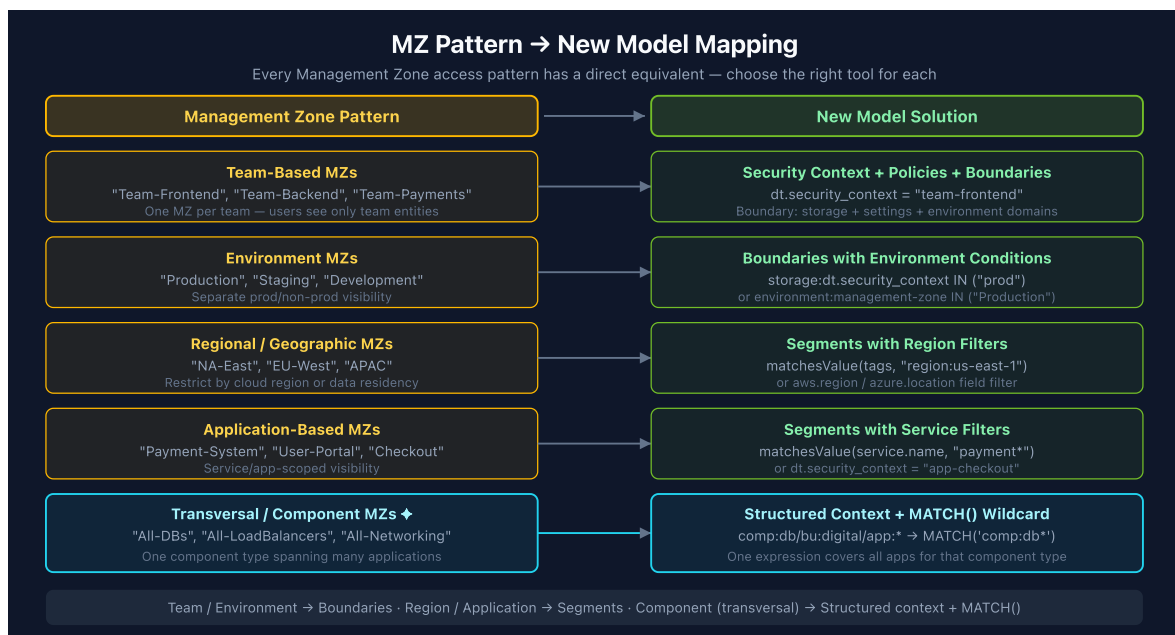
1. Navigate to **Group Management**
2. Select or create a group
3. Go to **Permissions** tab
4. Click **Add permission**
5. Select:
 - **Policy:** Choose policy to bind

- **Boundary** (optional): Select restriction
- **Environment**: Select target environment

6. Click **Save**

8. Mapping MZ Access to Policies + Boundaries

Common MZ Permission Patterns



MZ Permission	Equivalent Policy + Boundary
View only	Standard User + data-read policies + Boundary
View + Edit	Pro User + Settings Writer + Boundary
Full access to MZ	Pro User + All Grail data read access + Boundary
Admin in MZ	Admin User (or a custom policy) + Boundary

Migration Example: Team-Based MZ

Before (MZ-based):

- Management Zone: "Frontend-Team"
- Rules: Services with tag `team:frontend`
- Users assigned via RBAC

After (Policy + Boundary):

1. Set security context on entities:
 - Services get `dt.security_context = "team-frontend"`
2. Create two parallel policy bindings – Gen3 with Gen3 boundary, Gen2 with Gen2 boundary:

Binding 1 – Gen3 (canonical):

```
ALLOW storage:logs:read, storage:spans:read, storage:metrics:read,  
      settings:objects:read, settings:objects:write  
WHERE storage:dt.security_context IN ("team-frontend")  
      AND settings:dt.security_context IN ("team-frontend");  
(policy WHERE uses AND, not OR – each condition scopes its own service)
```


Binding 2 – Gen2 (transitional, until MZ retirement):

```
ALLOW environment:roles:viewer  
WHERE environment:management-zone IN ("Frontend-Team");
```
3. Attach both bindings to the group:
Group: Frontend Developers (SAML)
Bindings: Gen3 binding + Gen2 binding (both active during migration window)

Migration Example: Environment MZ

Before (MZ-based):

- Management Zone: "Production"
- Rules: Hosts with tag `env:production`
- Users get MZ-filtered view

After (Policy + Boundary):

```
1. Set security context on entities:
  - Hosts get dt.security_context = "prod-{region}"

2. Create two boundaries (don't mix Gen2 and Gen3 in one):
  Gen3 boundary "Production - Grail":
    storage:dt.security_context startsWith "prod-";
    settings:dt.security_context startsWith "prod-";
  Gen2 boundary "Production - Classic":
    environment:management-zone IN ("Production");

3. Bind to group with two parallel bindings:
  Group: Production Operators (SAML)
  Binding 1 (Gen3): Pro User + data-read policies, Boundary "Production - Grail"
  Binding 2 (Gen2): Environment role - Access environment, Boundary "Production - Classic"
```

9. Security Context and Bucket Strategy

What Is Security Context?

Security Context (`dt.security_context`) is an entity attribute used for access control. It's the recommended way to scope boundaries for Grail data.

Primary Grail Fields

When migrating from Management Zones, leverage **Primary Grail Fields** for consistent data partitioning:

Field	Source	MZ Migration Use
<code>k8s.cluster.name</code>	Kubernetes	Replace cluster-based MZs
<code>k8s.namespace.name</code>	Kubernetes	Replace namespace-based MZs
<code>aws.account.id</code>	AWS metadata	Replace AWS account MZs
<code>dt.host_group.id</code>	OneAgent config	Replace host group-based MZs

Bucket Strategy for Migration

Buckets partition Grail data and integrate with policies and boundaries for team-level access control.

Bucket Limitations

 **Critical:** Plan buckets carefully during migration - these constraints cannot be changed later.

Limitation	Details
One data type per bucket	Logs, metrics, events, OR spans - not mixed
Names are immutable	Bucket names CANNOT be changed after creation
No data migration	Data CANNOT be moved between buckets
Naming rules	3-100 chars, lowercase alphanumeric, underscores, hyphens only
Maximum buckets	80 per environment (default limit)
Optimal ingest	~1 TB/day per bucket for best query performance
Acceptable ingest	1-3 TB/day per bucket (limited query window)
Maximum ingest	3 TB/day per bucket hard limit

Query Constraints

Limit	Value	Impact
Maximum data scanned	500 GB	Limits queryable time window
Maximum records returned	1,000	Use aggregations for larger datasets
Maximum response payload	1 MB	Large result sets may be truncated

Default Bucket Retentions

Bucket	Retention
default_logs	35 days
default_metrics	15 months
default_spans	10 days
default_events	35 days

Using Buckets for Team Isolation (MZ Replacement)

When replacing team-based MZs, use buckets in **both policies AND boundaries**:

Pair Gen3 policies with Gen3 boundaries; pair Gen2 policies with Gen2 boundaries. Two parallel bindings on the group, not one mixed boundary:

```
# Gen3 policy
ALLOW storage:logs:read,
        storage:spans:read,
        storage:metrics:read,
        settings:objects:read,
        settings:objects:write;
```

```
# Gen3 boundary – scoped by bucket + security context
storage:bucket-name      IN ("frontend_logs", "frontend_spans",
"frontend_metrics");
storage:dt.security_context  IN ("team-frontend");
settings:dt.security_context IN ("team-frontend");
```

```
# Gen2 policy – Classic entity access, transitional
ALLOW environment:roles:viewer;
```

```
# Gen2 boundary
environment:management-zone IN ("Frontend-Team");
```

Don't use `MATCH` on Gen2 policies (e.g., `MATCH('*')` against `storage:entities:read` — that's a Gen2 surface). `MATCH` is a Gen3-only operator;

using it as a wildcard in a Gen2 boundary will silently grant access to all Gen2 entities.

Migration Mapping: MZ to Buckets

MZ Pattern	Bucket Strategy
Team MZ (e.g., "Frontend-Team")	Team-specific buckets: frontend_logs , frontend_spans
Environment MZ (e.g., "Production")	Environment buckets: prod_logs , prod_metrics
Regional MZ (e.g., "US-East")	Region buckets: useast_logs , useast_spans
Compliance MZ (e.g., "PCI")	Compliance buckets: pci_logs , pci_events

Complete Team Migration Example

Before (MZ-based):

- Management Zone: "Checkout-Team"
- Users see all data within the MZ

After (Policy + Bucket + Boundary):

```
Group: Checkout-Team (SAML from AD)
├─ Policy: Checkout Data Access
│   ├── ALLOW storage:logs:read WHERE storage:bucket-name = "checkout_logs"
│   ├── ALLOW storage:spans:read WHERE storage:bucket-name = "checkout_spans"
│   └─ ALLOW storage:metrics:read WHERE storage:bucket-name =
"checkout_metrics"
└─ Boundary:
    storage:bucket-name IN ("checkout_logs", "checkout_spans",
"checkout_metrics");
    storage:dt.security_context IN ("checkout");
    environment:management-zone IN ("Checkout-Team");
```

For detailed bucket design guidance, see **ORGNZ-03: Bucket Strategy and Design** which covers naming conventions, retention planning, and cost attribution patterns in depth.

Setting Security Context

Security context can be set via:

- **Entity Enrichment rules** (recommended)
- **Host properties** via OneAgent
- **OpenPipeline processing**
- **Auto-tagging rules** (legacy)

Security Context Naming Strategy

Pattern	Example	Use Case
team-{name}	team-frontend	Team ownership
env-{name}	env-production	Environment separation
region-{code}	region-us-east	Geographic isolation
app-{name}	app-checkout	Application scoping
{env}-{team}	prod-frontend	Combined scoping

For transversal access patterns (teams spanning multiple applications by component type — database, networking, OS), consider a structured multi-dimensional format: `comp:<component>/bu:<business-unit>/app:<application>` . This enables `MATCH('comp:db*')` to reach all database layers across all applications, while `MATCH('*/app:easytrade')` scopes to all components of one app. The dimension you need for transversal slicing must come *first* for Classic entity access. See **IAM-04: Policy Authoring** for the complete design pattern.

Query Entities by Security Context

```
// List services with their security context
// Identify services that need security context assignment
fetch dt.entity.service
| fields entity.name,
        dt.security_context,
        tags,
        managementZones
| sort dt.security_context asc
| limit 50

// Alternative: Smartscape on Grail (entity.name → name)
// smartscapeNodes SERVICE
// | fields name,
// dt.security_context,
// tags,
// managementZones
// | sort dt.security_context asc
// | limit 50
```

```
// Count entities by security context
// Helps verify security context distribution
fetch dt.entity.service
| summarize count = count(), by:{dt.security_context}
| sort count desc
```

10. Custom Policies

When to Create Custom Policies

- Default policies are too broad or narrow
- Need specific permission combinations
- Require conditional access
- Compliance requirements

Custom Policy Examples

Log Viewer for Specific Team:

```
// Policy Name: Frontend Team Log Access
// Description: Read-only access to frontend service logs

ALLOW storage:logs:read
  WHERE storage:dt.security_context = "team-frontend"

ALLOW storage:spans:read
  WHERE storage:dt.security_context = "team-frontend"
```

Environment-Restricted Admin:

```
// Policy Name: Development Admin
// Description: Full access to development environment only

ALLOW storage:*:*
  WHERE storage:dt.security_context startsWith "dev-"

ALLOW settings:objects:*
  WHERE settings:scope startsWith "environment:dev"
```

Dashboard and Notebook Creator:

```
// Policy Name: Dashboard Creator
// Description: Create and manage dashboards and notebooks

ALLOW document:documents:read
ALLOW document:documents:write
ALLOW document:documents:delete
ALLOW document:direct-shares:write

// Read data for dashboards
ALLOW storage:logs:read
ALLOW storage:metrics:read
ALLOW storage:spans:read
```

11. Best Practices

Do's

-  **Start with default policies** - customize only when needed
-  **Use least privilege** - grant minimum required permissions
-  **Document policies and boundaries** - clear names and descriptions
-  **Test before production** - verify with test users
-  **Use boundaries for scope** - keep policies reusable
-  **Include all three domains** in boundaries for complete MZ replacement
-  **Use SAML groups** tied to AD for team management

Don'ts

-  **Don't create policy per user** - use groups
-  **Don't duplicate default policies** - extend instead
-  **Don't use overly broad wildcards** - be specific
-  **Don't embed conditions in policies** when boundaries are better
-  **Don't forget the settings domain** in boundaries

Naming Conventions

```
// Good names
"Frontend Team - Standard Access"    (policy)
"Production Environment"             (boundary)
"L0B5-Scope"                        (boundary)

// Bad names
"Policy 1"
"Test"
"John's boundary"
```

Audit Checklist

- ☐ Each boundary has a clear purpose
- ☐ Boundary names follow naming convention
- ☐ All three domains included (environment, storage, settings)

- [] SAML groups aligned with AD structure
- [] Dependencies tracked (which policies use which boundaries)
- [] Regular review scheduled

12. Troubleshooting

Common Issues

Issue	Likely Cause	Solution
User can't access anything	No policy bound	Bind policy to user's group
User sees too much	Boundary too broad or missing	Tighten boundary conditions
User can't see expected data	Missing data policy	Add a data-read policy (e.g. <code>Read Logs / All Grail data read access</code>)
Permissions inconsistent	Multiple conflicting policies	Review all bound policies
Works in classic, not Grail	Only environment domain used	Add storage domain (Gen3 boundary)
Settings not restricted	Missing settings domain	Add <code>settings:dt.security_context</code> boundary

Debugging Steps

1. **Verify group membership:** Is user in correct group?
2. **Check policy bindings:** What policies are bound to group?
3. **Review boundaries:** Are both Gen3 and Gen2 boundaries attached where needed?
4. **Test with admin:** Does admin see the data?
5. **Check security context:** Is it set on the entities?

Testing Boundaries

1. **Create test user** in target group
 2. **Log in as test user**
 3. **Verify access:**
 - Can access expected data?
 - Cannot access restricted data?
 4. **Test edge cases:**
 - Entities at boundary edges
 - New entities without security context
-

Summary

In this notebook, you learned:

1. **Policies define WHAT**, boundaries define **WHERE**
2. **Default policies** cover most use cases - customize only when needed
3. **Shared syntax** for conditions in both policies and boundaries
4. **Three domains** needed for complete MZ replacement (environment, storage, settings)
5. **SAML + Policy + Boundary** structure for AD integration
6. **Migration patterns** for team-based and environment-based MZs

Next Steps

Continue to **MZ2POL-05: Segments Implementation** to:

- Create DQL-based data filters
- Replace MZ data filtering with Segments
- Configure cross-app filtering

For large-scale migrations with many Management Zones, see **MZ2POL-08:**

Templated Policies for MZ Migration to:

- Convert MZ patterns to parameterized policy templates
- Automate bulk policy binding via IAM API
- Validate migrated access with DQL queries

Additional Resources

- [Working with Policies](#)
 - [IAM Policy Reference](#)
 - [Policy Boundaries Documentation](#)
 - [Grant Access with Security Context](#)
 - [Policy Templating](#)
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